

Micro-Mouse Design Report

Please note that you are required to use the Microsoft form when preparing your report: it imposes strict formatting that must not be changed and enforces character limits for each section. **Do not disable the “Protect Form” functionality in Microsoft Word.** After completion you should convert the document to PDF for final submission.

If the form misbehaves then prepare your content in a separate application, then use copy and paste to populate it.

Student number: MNLIMT001

Brief description of design task (max. 600 characters)

A short outline of the task identified and the problem it is intended to solve.

The design task chosen was to program the micro mouse (MM) to navigate a dead end using Time-of-Flight (ToF) sensors. This involves enabling the MM to detect when it has reached a dead end and execute a turn-around manoeuvre to continue navigating the maze.

The main challenge lies in accurately interpreting ToF sensor readings to identify dead ends and implementing a reliable control algorithm for smooth, precise turning. Successfully solving this task ensures consistent navigation of corners and straight paths, forming the foundation for full maze exploration which is the main design goal.

Visual aids for this report

Any figures, images, or other graphics can be placed in this section and be referred to in the rest of the text. This can include excerpts of Stateflow or Simulink block diagrams. Note that you can only insert a single graphic file here, so prepare your content separately, convert to an image, and insert as required.

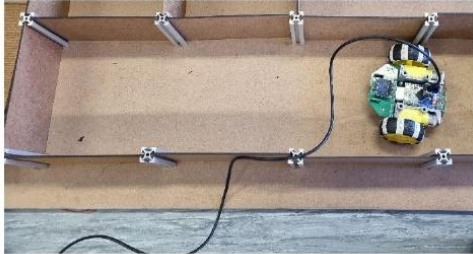


Figure 1: The Micromouse positioned at the entrance of a dead end in the maze. The robot is about to navigate the dead end using its forward, left, and right TOF sensors. Sensor readings and motor actions are captured in real time using Rapid Development, allowing data collection for wall detection, turn initiation, and encoder-based 180° turn analysis. This setup illustrates the MM's starting alignment, sensor coverage, and the controlled environment used to evaluate the dead-end navigation strategy.

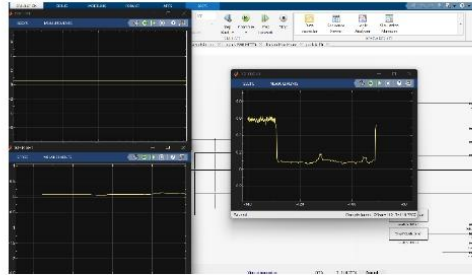


Figure 2: Rapid Development scope output showing TOF sensor readings as the Micromouse navigates a dead end. The front sensor initially measures high distances (~0.6 m) while approaching the wall, then drops sharply to ~0.1 m as the wall is detected, with some expected noise. After executing the 180° turn, the front sensor reading rises again (~0.5 m) as no wall is present ahead. The left and right sensors remain approximately constant near zero throughout the process, reflecting the presence of walls on either side of the MM. The transition between detection, turn, and resumption of forward motion appears smooth, demonstrating effective real-time data capture and control using Rapid Development.

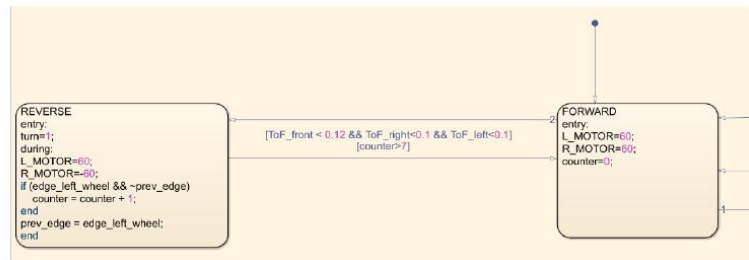


Figure 3: Stateflow model in Simulink for the Micromouse (MM) to follow a dead-end navigation trajectory. The model contains two main states: FORWARD, where the MM moves straight with wheel motors set for forward motion, and REVERSE (turn-around), where the left and right motors are set to opposite speeds to perform a 180° turn. Transitions between states are triggered by TOF sensor thresholds: when the front sensor detects a wall at <0.12 m and both side sensors detect walls (<0.1 m), the model enters the REVERSE state. Encoder tick counts are used to measure rotation, with a counter incremented on each detected left wheel edge. The transition back to FORWARD occurs when the counter exceeds 7, completing the turn.

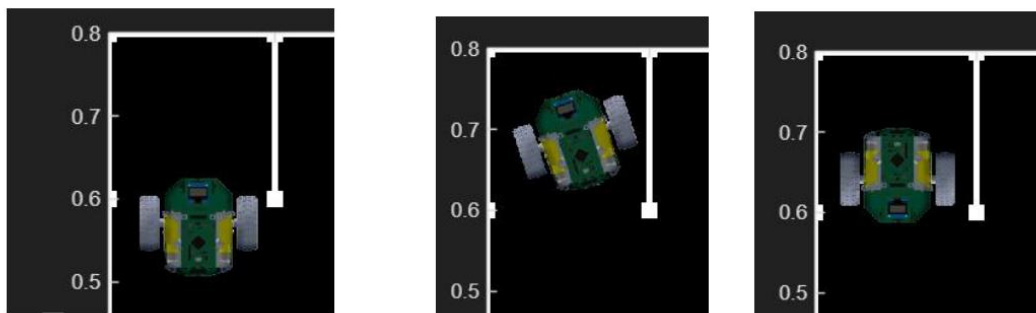


Figure 4: Simulation snapshots showing the Micromouse (MM) navigating a dead end. The first image depicts the MM approaching the dead end, with TOF sensors detecting walls in front and on the sides. The second image shows the MM executing a 180° turn, using encoder-based wheel rotations to turn around while maintaining smooth motion. The final image illustrates the MM moving away from the dead end, resuming forward navigation in the maze. These images demonstrate how the MM responds to sensor inputs in simulation, successfully detecting and exiting a dead-end scenario.

Design criteria and constraints (max. 1000 characters)

These are fundamental elements in the engineering design process. Criteria are the desired features, functions, and performance standards that a design must meet, while constraints are the limitations or restrictions that must be considered during the design process.

Criteria:

1. Detect dead-end when all TOF sensors ≤ 0.12 m.
2. Initiate turn-around by adjusting wheel direction.
3. Use encoder ticks to stop turn at 180° rotation.
4. Ensure smooth, stable, repeatable turns.
5. Resume forward motion accurately post-turn.
6. Filter sensor noise and debounce to prevent false triggers.

Constraints:

1. ToF sensor latency and noise require filtering.
2. Encoder resolution limited to 8 ticks/rev and MM hardware resolution.
3. DC motor nonlinearity (dead zones, torque limits).
4. Accuracy depends on wheel marker contrast and placement.
5. Battery goes below threshold needs manual jumpstart.
6. Environmental factors such as ambient lighting may affect readings.

Main design assumptions with justification (max. 1000 characters)

Design assumptions are fundamental beliefs or statements about a system, its environment, or its users, that are taken as true without proof during the design process. Justification involves providing reasons and evidence to support these assumptions, explaining why they are reasonable and necessary for the design to proceed.

Design Assumptions:

1. TOF sensors (VL53L1X) provide accurate distance readings. This was verified in real-life tests by placing objects at known distances and comparing the sensor readings.
2. The physical maze matches the simulated maze in width, allowing sufficient space for 180° turns.
3. The MM operates on a flat, obstacle-free surface, justified as testing occurs in controlled lab conditions.
4. Wheel slippage is negligible due to high-friction wheels and smooth surfaces.
5. Motors deliver consistent performance, as the MM is regularly charged.
6. The MM moves reliably in a straight line and self-corrects, as shown in Milestone 1.
7. The initial alignment of the MM before each run is assumed correct, allowing symmetrical wall detection by all TOF sensors.
8. The MM cannot safely reverse due to its half-sphere support, making rotation the preferred maneuver.

Design process description (max. 1000 characters)

The design process is a structured approach, often iterative, used to solve problems and develop solutions. It typically involves identifying a need, researching the problem, brainstorming ideas, creating prototypes, testing, and refining the design until it meets the desired outcome.

The design process followed a structured, iterative approach to ensure a reliable and efficient solution. ToF sensors were tested in simulation and real life (Figures 1 and 4) to determine optimal thresholds. Initially, a front sensor threshold of 10 mm was used but proved too close for efficient turning, a final value of 0.12m for front and 0.1 m for left and right was chosen. The solution was modelled in Stateflow in Simulink (Figure 2) to manage the dead-end navigation.

For the turn, encoder counters were iteratively tuned until an exact 180° rotation was achieved. The left wheel encoder was used, as it provided clearer rising and falling edges than the right in real-life tests, while in simulation both behaved identically. The wheels were marked with evenly spaced tape, giving 8 ticks per revolution for accurate encoder readings. Encoders were chosen over the gyroscope due to superior repeatability and lower noise. Iterations also refined wheel speeds and sensor thresholds.

Design implementation (max. 1000 characters)

Design implementation describes the process of translating a design into a functional reality. It's the stage where the theoretical design is brought to life, whether it's a software application, a physical product, or a system.

The final design implemented the solution in Simulink with a Stateflow chart (Figure 3). The MM has two states: FORWARD and REVERSE. The inputs are ToF distance readings from front, left, and right sensors of the MM, and encoder ticks from the left wheel. Encoder ticks are calculated using a detect rise positive block to detect rising edges, which was done in Milestone 1. The straight-line correction logic from Milestone 1 was used to maintain accurate forward motion.

In the FORWARD state, both motors drive forward at a speed of 60, and the counter is reset to 0. A transition to the REVERSE state occurs when all three ToF sensors detect walls within threshold distances (front < 0.12 m, left and right < 0.1 m). In the REVERSE state, the left and right motors are set to opposite speeds (L=60, R=-60) to execute a 180° turn, while encoder ticks are counted. The MM returns to FORWARD once the counter reaches 7 ticks, completing the turn and allowing continued maze navigation.

Design evaluation and testing (max. 1000 characters)

Evaluation and testing is crucial for ensuring a design meets its objectives and user needs. This process involves assessing a design's feasibility, desirability, and viability, and identifying areas for improvement before it gets deployed.

The simulation results (Figure 4) demonstrated reliable dead-end detection at 0.12 m front and 0.1 m side thresholds, with 180° turns completed in 7 encoder ticks. Rapid Development testing (Figure 2) provided real-time data. Real-life tests (Figures 1 & 2) confirmed functionality, though front sensor readings were noisier, requiring a moving-average filter.

Deviations arose from maze misalignment and metal poles, which added approximately 1 cm to the effective width of the dead-end, further tuning to prevent collisions was needed. Encoder reliability was occasionally affected by dirt accumulation on the wheel tape, which caused misread or missed ticks; performance improved after replacing the contaminated tape. Low battery levels affecting motor performance. Overall, dead-end navigation was efficient, validating ToF thresholds, encoder counts, and wheel speeds.

Instructions

This report should document a structured design process that you conducted in relation to the micro-mouse project. You should explicitly be demonstrating use of methods presented to you in EEE3088F. The main purpose is to provide sufficient evidence for us to claim your proficiency in the ECSA Design Graduate Attribute (GA3), which is required to pass the course and complete your BSc(Eng) programme.

You can choose any design task for which you can confidently present evidence of design thinking. Without being prescriptive, some examples might be to design, build, and test:

1. a more realistic and operationally useful Simulink model of the micro-mouse motor drive system
2. a realistic Simulink model for time-of-flight distance measurements that function specifically in a maze environment
3. Simulink model enhancements that accommodate a realistic perturbation such as friction, wheel slip, variable robot load (mass), operation on an inclined plane, or other
4. processes for signal conditioning on measurements (e.g. improving IMU performance through calibration and noise modelling)
5. a process for smoothing robot hardware motion (e.g. prediction and filtering of actuation signals)
6. a Kalman filter-type algorithm for combining measurements and actuator signals in a structured manner
7. a coordinated turn procedure that permits 90 degree turns of the robot even with significant forward velocity
8. a method to repurpose the line-following sensors to function as rotary encoders of the wheels
9. a method for making the robot follow a predetermined trajectory
10. an overall improved micro-mouse simulator for in-the-loop algorithm development.

The task for which you provide evidence need not align with any submission milestone, and it does not have to be novel or earth shattering. However, the approach taken and the reporting thereof must follow formal design processes at the level required and must be objectively evaluated.

In essence, you should think of a micro-mouse related problem, approach it in a design-oriented manner, implement and test a proposed solution, and document the process you followed and the outcomes.

The format for this design report is very constrained. You are given limited space to present information relevant to each evaluation criterion, and each section will be evaluated for evidence it provides of your design thinking skills. A guide to the purpose of each section is as follows:

- **Brief description of design task:** A short outline of the task identified and the problem it is intended to solve.
- **Visual aids for this report:** Any figures, images, or other graphics can be placed in this section and be referred to in the rest of the text. This can include excerpts of Stateflow or Simulink block diagrams.
- **Design criteria and constraints:** These are fundamental elements in the engineering design process. Criteria are the desired features, functions, and performance standards that a design must meet, while constraints are the limitations or restrictions that must be considered during the design process.

- **Main design assumptions with justification:** Design assumptions are fundamental beliefs or statements about a system, its environment, or its users, that are taken as true without proof during the design process. Justification involves providing reasons and evidence to support these assumptions, explaining why they are reasonable and necessary for the design to proceed.
- **Design process description:** The design process is a structured approach, often iterative, used to solve problems and develop solutions. It typically involves identifying a need, researching the problem, brainstorming ideas, creating prototypes, testing, and refining the design until it meets the desired outcome.
- **Design implementation:** Design implementation describes the process of translating a design into a functional reality. It's the stage where the theoretical design is brought to life, whether it's a software application, a physical product, or a system.
- **Design evaluation and testing:** Evaluation and testing is crucial for ensuring a design meets its objectives and user needs. This process involves assessing a design's feasibility, desirability, and viability, and identifying areas for improvement before it gets deployed.

The marking rubric for these reports is as follows:

Topic (weight)	Unacceptable (0)	Marginal (1)	Acceptable (2)	Exceptional (3)
Visual aids for this report (3)	Visual aids are unclear, confusing, or difficult to understand. Fails to effectively communicate design elements due to poor visual presentation	Visual aids are somewhat effective, but unclear or confusing in parts. May lead to misinterpretation of design elements due to poor visual presentation	Visual aids are clear and effectively communicate design elements. Meets minimum requirements for visual clarity and presentation	Visual aids are highly effective, clear, and engaging. Exceeds expectations for visual communication skills, creativity, and presentation
Design criteria and constraints (3)	Fails to identify or acknowledge critical design constraints. Ignores fundamental principles that limit design	Partially identifies some, but not all, critical constraints. Assumes factors without thorough consideration of limitations	Clearly identifies and justifies relevant design constraints. Demonstrates consideration of limitations or trade-offs in design decisions	Thoroughly identifies all critical constraints. Consistently considers design principles that account for limitations and trade-offs
Main design assumptions with justification (3)	Assumptions not identified or justified. Ignores critical variables and constraints	Lacks clear identification or justification of assumptions. Assumptions not thoroughly	Assumptions clearly identified and justified. Consistently applied problem constraints with	Assumptions rigorously identified and justified through thorough analysis. Assumptions

		explored and considered	evidence to support	explicitly made explicit and transparent
Design process description (3)	Lacks clear description of design process. Fails to demonstrate understanding of design methodology	Partially describes design process but lacks detail. Fails to identify key steps or decisions in the design process	Clearly describes main stages of the design process. Demonstrates understanding of key design concepts and principles	Comprehensive description of the entire design process. Clearly explains thought processes, decisions, and trade-offs
Design implementation (3)	Does not demonstrate successful implementation of design. Fails to demonstrate functional or effective design solution	Partially implements design, but with significant flaws. Fails to demonstrate clear and consistent implementation of design	Demonstrates successful implementation of basic design elements. Shows clear and consistent execution of design principles	Seamlessly implements design elements with attention to detail. Demonstrates exceptional execution of design principles
Design evaluation and testing (3)	Ignores or fails to evaluate design thoroughly. Fails to test and validate design assumptions	Partially evaluates design, but with significant limitations. Fails to test and validate design assumptions, or tests are not comprehensive	Consistently evaluates design for functionality and usability. Tests and validates design assumptions	Thoroughly evaluates design for all aspects. Validates design assumptions through rigorous testing and analysis

A discretionary further 2 points will be awarded for overall quality of the work and coherence of the report.

Note that to satisfy the GA assessment you will be required to get at least 1.5 marks out of 3 for every topic category in the rubric (i.e. no “Unacceptable” anywhere in the report, and “Marginal” is only permitted if an additional 0.5 marks is added via the rubric). This is separate from the mark that you receive, and a passing mark does not indicate that you’ve successfully attained the relevant GA.